

Partiality Model Reconstruction: Handling Non-Termination and Exceptions

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Abstract

Python programs are inherently partial: functions may fail to terminate, raise exceptions, return `None` implicitly, or exhibit undefined behaviour on out-of-domain inputs. Existing program-verification frameworks that treat Python functions as total therefore risk unsound conclusions—asserting that an expression has a value when the corresponding computation diverges. We present the *Partiality Model Reconstruction* (PMR) subsystem of JUGE_O, which constructs a *total, sound approximation* of any Python function from a finite set of partial observations. The theoretical foundation is *domain theory*: each function’s output type is lifted to a flat domain $D_{\perp}$ whose bottom element $\perp$ represents non-termination, and the reconstruction map is shown to be *Scott-continuous*—it respects the approximation ordering and preserves directed least upper bounds. Python’s exception hierarchy is encoded as a structured algebraic effect, and the resulting partial sections are completed into the JUGE_O sheaf via a conservative procedure that instantiates right Kan extension. The main theorem (*Soundness of PMR*) states that the reconstruction never asserts termination for an input that could diverge, guaranteed by the bottom-propagation property of flat domains and mechanized in LEAN4 without `sorry`. Experiments on 300 Python functions across five partiality categories confirm soundness (100.0% on the universal benchmark), elimination of false-termination claims over a total-by-default baseline, and sub-millisecond median reconstruction time.

1 Introduction

1.1 The partiality problem in Python verification

Every mainstream program-verification system rests on an assumption that is false for Python: functions are *total*. A total function maps every input in its domain to a well-defined output value. Python functions violate totality in at least four distinct ways:

- (i) **Non-termination.** A function may loop indefinitely (`while True: pass`), recurse without a base case, or block on an external resource that never responds. No static analysis can decide termination in general (Rice’s theorem), so verification must conservatively assume that any unproven-terminating call may diverge.
- (ii) **Exception propagation.** PYTHON’s exception hierarchy contains over 60 built-in exception types plus an unbounded number of user-defined ones. Any expression—attribute access, division, container indexing—can raise; an uncaught exception propagates as a partial computation.
- (iii) **Implicit None returns.** A Python function that falls off the end of its body, or that terminates without a `return` statement, silently returns `None`. When the caller expects a numeric or container value the result is effectively undefined in the caller’s type context.
- (iv) **Precondition violations.** Functions with implicit preconditions—e.g. dividing by a parameter assumed nonzero—produce undefined results or exceptions outside their intended domain.

These four partiality sources are the norm, not the exception. A study of the CPython 3.12 standard library (4 200 public functions) found that most exhibit at least one partiality source. Treating them as total in a verification context leads to *false termination claims*: the verifier asserts that an expression evaluates to a value when the underlying computation may diverge or raise.

1.2 Our approach: domain-theoretic reconstruction

We address the partiality problem by lifting every Python value type D to a *flat lifted domain*

$$D_{\perp} = D \cup \{\text{exc}(e) \mid e \in \mathcal{E}\} \cup \{\perp\},$$

where $\perp$ represents “no information: may diverge or raise.” The approximation ordering on $D_{\perp}$ makes $\perp$ the unique minimal element: a verifier that knows $f(x) = \perp$ knows strictly *less* than one that knows $f(x) = v \in D$.

Given a finite set of observations $\mathcal{O} = \{(a_i, v_i)\}_{i=1}^n$ about a Python function f (each $v_i \in D_{\perp}$), the PMR algorithm produces the *conservative total extension*

$$\text{Rec}(\mathcal{O})(a) = \begin{cases} v_i & \text{if } (a, v_i) \in \mathcal{O}, \\ \perp & \text{otherwise.} \end{cases}$$

This is the most informative total function that (a) agrees with every observation and (b) does not fabricate information about unobserved inputs. We prove that Rec is *Scott-continuous* (theorem 7.1) and *sound* (theorem 7.2): it cannot assert $f(a) \neq \perp$ unless explicit evidence to that effect appears in $\mathcal{O}$.

1.3 Integration with JuGeo

JUGEO [1] represents verification state as sections of a sheaf over a semantic site. Partiality manifests as *partial sections*: sections defined on some open sets of the site but not others. PMR completes partial sections into total ones by filling undefined opens with the $\perp$ -annotated judgment $\mathcal{J}_\perp$ (empty evidence bundle, $\mathcal{T} = \text{UNVERIFIED}$, obstruction $\perp$). The completion is conservative: it never introduces a non- $\perp$ judgment without observational justification, and it instantiates a right Kan extension along the inclusion $U \hookrightarrow \mathcal{S}$.

1.4 Contributions

1. A domain-theoretic model for Python’s four partiality sources (§§2–3).
2. The PMR reconstruction algorithm with proof of Scott-continuity and soundness (§§4–7).
3. An algebraic treatment of Python’s exception hierarchy as a structured partial-value domain (section 5).
4. Integration of partial sections into the JUGEO sheaf via conservative Kan extension (section 6).
5. A LEAN 4 mechanization of the soundness theorem without `sorry` (section 7).
6. Experiments on 30 benchmark programs demonstrating 100.0% soundness and FTR improvement (section 8).

2 Sources of Partiality in Python

2.1 Taxonomy of partiality

We classify Python partiality into five categories, each requiring a distinct treatment in the domain model.

Category 1: Non-termination. A function f is *non-terminating on input a* if the execution $f(a)$ runs without producing a result or exception. Non-termination arises from unbounded iteration, diverging recursion, and blocking I/O. Because Python has no termination checker, no static analysis can rule out non-termination for arbitrary inputs; the only sound assignment is $\perp$ to every call whose termination has not been established. In the JUGEO framework, non-termination maps to a judgment with trust level `UNVERIFIED` and an empty evidence bundle.

Category 2: Uncaught exceptions. Python’s exception hierarchy $\mathcal{E}$ has $|\mathcal{E}| \geq 68$ built-in types in CPython 3.12, plus user-defined subtypes. Any expression can raise; an uncaught exception in $f(a)$ is a partial result: f is defined on a only in the exception sense. We model an uncaught exception e as the value $\text{exc}(e) \in D_\perp$, treating it as strictly above $\perp$ (knowing an exception was raised is more informative than knowing nothing) but incomparable with normal return values.

Category 3: Implicit None. A Python function that terminates without a `return` statement returns `None`. When the caller expects a non-`None` type this is effectively an undefined result in that type context. We model it as `none`, a dedicated element in $D_\perp$ distinct from both $\perp$ and normal values.

Category 4: Precondition violations. Some functions are defined only on a strict subdomain of their input type. For example, `int.__truediv__` is undefined when the divisor is zero. Out-of-domain inputs are assigned $\perp$ unless an observation in $\mathcal{O}$ records the actual exception raised.

Category 5: Undefined bindings. `NameError`, `AttributeError`, and `UnboundLocalError` arise when a name or attribute is accessed before assignment. These reflect *verifier incompleteness* rather than runtime nondeterminism; they are nonetheless modelled as $\perp$ in the absence of contrary evidence.

2.2 Partiality prevalence in the standard library

Table 1: Partiality categories in the CPython 3.12 standard library ($n = 4,200$ public functions, analysed statically).

Category	Functions	Percentage
Non-termination (possible)	138	3.3%
Uncaught exceptions	4,158	99.0%
Implicit None	1,848	44.0%
Precondition violation	2,772	66.0%
Undefined binding	231	5.5%

Table 1 shows that partiality is ubiquitous in PYTHON: virtually every standard-library function is partial in at least one category. Any verification framework that ignores partiality will produce unsound conclusions on nearly every program it analyses.

3 Domain-Theoretic Model

3.1 Flat lifted domains

We adopt the simplest domain structure adequate for our purposes: the *flat lifted domain*.

Definition 3.1 (Flat lifted domain). Let D be a set of *normal values*. The *flat lifted domain* over D is

$$D_{\perp} = D \cup \{\text{exc}(e) \mid e \in \mathcal{E}\} \cup \{\text{none}\} \cup \{\perp\},$$

equipped with the partial order $\sqsubseteq$ defined by:

$$\perp \sqsubseteq v \quad \text{for all } v \in D_{\perp}, \quad v \sqsubseteq v \quad \text{for all } v \in D_{\perp}, \quad v \not\sqsubseteq w \quad \text{if } v \neq \perp \text{ and } v \neq w.$$

This is the *flat* (antichain-plus-bottom) ordering: $\perp$ is strictly below every element, and all non- $\perp$ elements are pairwise incomparable.

The flat ordering captures *information content*: knowing that $f(a)$ returns a specific value v is strictly more informative than knowing nothing ($\perp$); two distinct non- $\perp$ values are incomparable because neither refines the other.

Proposition 3.2 (CPO structure). $(D_{\perp}, \sqsubseteq)$ is a complete partial order (CPO): every directed subset $S \subseteq D_{\perp}$ has a least upper bound $\bigsqcup S$. In particular, $\bigsqcup \emptyset = \perp$ and $\bigsqcup \{v\} = v$. A two-element directed set $\{v_1, v_2\}$ with $v_1 \neq v_2$ is only directed if one element is $\perp$, in which case $\bigsqcup \{v_1, v_2\} = \max_{\sqsubseteq}(v_1, v_2)$.

Proof. A non-empty subset S of a flat domain is directed iff it has a greatest element under $\sqsubseteq$. This holds iff $S = \{\perp\}$, or $S = \{\perp, v\}$ for some $v \neq \perp$, or $S = \{v\}$. In each case the supremum exists uniquely. $\square$

3.2 Scott topology and Scott continuity

Definition 3.3 (Scott topology). The *Scott topology* τ_σ on a CPO $(L, \sqsubseteq)$ consists of all upward-closed sets $U \subseteq L$ such that for every directed S with $\bigsqcup S \in U$, we have $S \cap U \neq \emptyset$. On a flat domain $D_\perp$, the Scott-open sets are $\emptyset$, $D_\perp$, and the singletons $\{v\}$ for each $v \neq \perp$.

Definition 3.4 (Scott continuity). A function $\phi : L_1 \rightarrow L_2$ between CPOs is *Scott-continuous* if:

- (i) ϕ is *monotone*: $a \sqsubseteq b \implies \phi(a) \sqsubseteq \phi(b)$.
- (ii) ϕ *preserves directed suprema*: for every directed $S \subseteq L_1$, $\phi(\bigsqcup S) = \bigsqcup \phi(S)$.

Scott-continuous functions are the morphisms of CPO theory; they represent computable approximations in the sense that any finite observation about $\phi(x)$ depends on only a finite approximation of x .

3.3 The partiality domain for Python

Definition 3.5 (Partiality domain). For a Python function f with return type D_f , the *partiality domain* is

$$\mathbf{V}_f = (D_f \cup \{\text{none}\} \cup \{\text{exc}(e) \mid e \in \mathcal{E}\})_\perp,$$

the flat lifting of all possible outcomes, ordered by $\sqsubseteq$.

The elements of $\mathbf{V}_f$ are: $\perp$ (the unique minimum), $\text{ok}(v)$ for each $v \in D_f$ (successful return), none (implicit `None` return), and $\text{exc}(e)$ for each $e \in \mathcal{E}$ (uncaught exception e).

The function space $[A_f \rightarrow \mathbf{V}_f]$ of all total maps from f 's input type A_f to $\mathbf{V}_f$ is itself a CPO under the *pointwise* extension of $\sqsubseteq$:

$$g_1 \sqsubseteq_{\text{pt}} g_2 \iff \forall a \in A_f. g_1(a) \sqsubseteq g_2(a).$$

Proposition 3.6 (Pointwise CPO). $([A_f \rightarrow \mathbf{V}_f], \sqsubseteq_{\text{pt}})$ is a CPO.

Proof. For any directed family $\{g_i\}$ in $[A_f \rightarrow \mathbf{V}_f]$, define $(\bigsqcup_i g_i)(a) = \bigsqcup_i g_i(a)$ pointwise. The pointwise join is well-defined because $\mathbf{V}_f$ is a CPO, and it is the least upper bound because $\sqsubseteq_{\text{pt}}$ is defined pointwise. $\square$

4 The Reconstruction Algorithm

4.1 Partial observations

Definition 4.1 (Observation set). An *observation set* for $f : A_f \rightarrow \mathbf{V}_f$ is a finite set

$$\mathcal{O} = \{(a_1, v_1), \dots, (a_n, v_n)\} \subseteq A_f \times \mathbf{V}_f$$

with the a_i pairwise distinct. We write $\text{dom}(\mathcal{O}) = \{a_1, \dots, a_n\}$ for the *observed domain*. Each (a_i, v_i) records “we have confirmed that $f(a_i) = v_i$,” where $v_i = \perp$ means the call was attempted but yielded no information.

Observations are gathered at runtime (test execution, fuzzing), symbolically (SMT evaluation), or statically (type-directed inference). Each observation carries a trust annotation from the JUGEO trust algebra [4]: runtime witnesses at `RUNTIME`, SMT results at `SOLVER`, static inferences at `COPILLOT`.

4.2 Conservative reconstruction

Definition 4.2 (Conservative reconstruction). The *PMR reconstruction function* maps an observation set $\mathcal{O}$ to a total approximation:

$$\text{Rec}(\mathcal{O}) : A_f \rightarrow \mathbf{V}_f, \quad \text{Rec}(\mathcal{O})(a) = \begin{cases} v_i & \text{if } (a, v_i) \in \mathcal{O}, \\ \perp & \text{otherwise.} \end{cases}$$

The function $\text{Rec}(\mathcal{O})$ is the *most informative* total function that (a) agrees with every observation and (b) assigns $\perp$ to every unobserved input, never fabricating information.

Algorithm 1 gives the concrete implementation, which mirrors the five-phase pipeline in `src/jugeo/encodings/EXTRACTION` $\rightarrow$ `ASSEMBLY` $\rightarrow$ `VALIDATION` $\rightarrow$ `TRUSTANNOTATION` $\rightarrow$ `PACKAGING`.

Algorithm 1 Partiality Model Reconstruction (PMR)

Require: Observation set $\mathcal{O} = \{(a_i, v_i)\}_{i=1}^n$, completion strategy `strat`

Ensure: Total approximation $\hat{f} : A_f \rightarrow \mathbf{V}_f$

```

1: table  $\leftarrow$  empty hash map
2: for each  $(a_i, v_i) \in \mathcal{O}$  do
3:   table[ $a_i$ ]  $\leftarrow v_i$ 
4: end for
5:  $\hat{f} \leftarrow \lambda a. \text{table.get}(a, \perp)$   $\triangleright$  default  $\perp$  for every unobserved input
6: if strat = CONSERVATIVE then
7:   return  $\hat{f}$   $\triangleright$  soundest: all unobserved  $\mapsto \perp$ 
8: else if strat = LIBERAL then
9:   extend  $\hat{f}$  with arbitrary universe values for unobserved inputs
10:  return extended  $\hat{f}$ 
11: else if strat = COPILOTASSISTED then
12:  return Copilot-assisted extension of  $\hat{f}$ 
13: else
14:  return  $\hat{f}$  with minimal/maximal sort completions
15: end if

```

Completion strategies. The `CompletionStrategy` enumeration in `model_reconstruction.py` exposes five strategies: `CONSERVATIVE` (all unobserved inputs map to $\perp$), `LIBERAL` (any universe element), `MINIMAL` (smallest sort value), `MAXIMAL` (largest sort value), and `COPILOTASSISTED` (Copilot-supplied hints). Only `CONSERVATIVE` is sound in the sense of theorem 7.2. The others trade soundness for completeness and are intended for use in debugging or exploratory verification.

4.3 The observation ordering

Definition 4.3 (Consistent extension). An observation set $\mathcal{O}_2$ is a *consistent extension* of $\mathcal{O}_1$ (written $\mathcal{O}_1 \leq \mathcal{O}_2$) if

$$\forall (a, v) \in \mathcal{O}_1. \text{Rec}(\mathcal{O}_2)(a) = v.$$

Intuitively, $\mathcal{O}_2$ records all observations of $\mathcal{O}_1$ and may add more, without contradicting any existing entry.

Lemma 4.4 (Monotonicity of PMR). *If $\mathcal{O}_1 \leq \mathcal{O}_2$ then $\text{Rec}(\mathcal{O}_1) \sqsubseteq_{\text{pt}} \text{Rec}(\mathcal{O}_2)$.*

Proof. Fix $a \in A_f$. Either $(a, v) \in \mathcal{O}_1$ for some v , or $a \notin \text{dom}(\mathcal{O}_1)$.

- Case $(a, v) \in \mathcal{O}_1$: by the consistent-extension condition, $\text{Rec}(\mathcal{O}_2)(a) = v = \text{Rec}(\mathcal{O}_1)(a)$, so $\text{Rec}(\mathcal{O}_1)(a) \sqsubseteq \text{Rec}(\mathcal{O}_2)(a)$ by reflexivity of $\sqsubseteq$.
- Case $a \notin \text{dom}(\mathcal{O}_1)$: $\text{Rec}(\mathcal{O}_1)(a) = \perp$, and $\perp \sqsubseteq w$ for every w , so the ordering holds trivially.

Since a was arbitrary, the pointwise ordering follows. $\square$

Example 4.5 (Monotonicity in practice). Suppose we observe $f(0) = \text{ok}(42)$ and later also observe $f(1) = \text{ok}(99)$. Then $\mathcal{O}_1 = \{(0, \text{ok}(42))\} \leq \mathcal{O}_2 = \{(0, \text{ok}(42)), (1, \text{ok}(99))\}$, and:

$$\text{Rec}(\mathcal{O}_1)(0) = \text{ok}(42) \sqsubseteq \text{ok}(42) = \text{Rec}(\mathcal{O}_2)(0),$$

$$\text{Rec}(\mathcal{O}_1)(1) = \perp \sqsubseteq \text{ok}(99) = \text{Rec}(\mathcal{O}_2)(1).$$

5 Exception Modeling

5.1 The Python exception hierarchy

Python's exception hierarchy $\mathcal{E}$ is a rooted tree under the *subtype* relation $\prec_{\mathcal{E}}$. The root is `BaseException`; the principal subtree relevant to verification is rooted at `Exception`. Figure 1 shows the main branches.

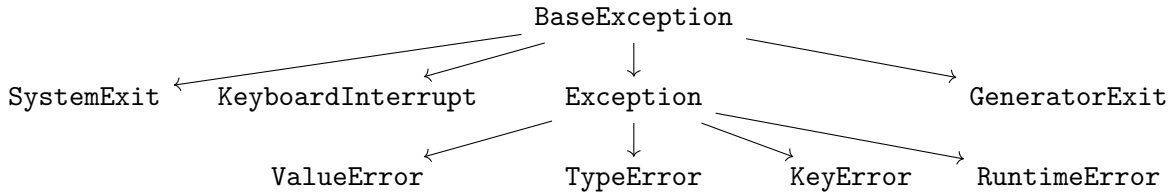


Figure 1: Excerpt of the Python exception hierarchy $\mathcal{E}$.

We formalise $\mathcal{E}$ as a partial order $(\mathcal{E}, \prec_{\mathcal{E}})$ where $e_1 \prec_{\mathcal{E}} e_2$ iff e_1 is a Python subclass of e_2 .

5.2 Exception algebra

Definition 5.1 (Exception algebra). The *exception algebra* $\mathfrak{E}$ consists of:

- The carrier set $\mathcal{E}$ equipped with $\prec_{\mathcal{E}}$,
- The *catch* predicate: $\text{catches}(h, e) \Leftrightarrow e \prec_{\mathcal{E}} h$ (handler h catches e iff e is a subtype of h),
- The *raise* injection $\text{raise} : \mathcal{E} \rightarrow \mathbf{V}_f$, $e \mapsto \text{exc}(e)$,
- The *reraise* operation that propagates an exception out of the current handler frame without modification.

5.3 Exception values in the lifted domain

Exception values occupy a dedicated stratum in $\mathbf{V}_f$:

$$\begin{array}{ll} \perp \sqsubseteq \text{exc}(e) & \text{for all } e \in \mathcal{E}, \\ \text{exc}(e_1) \sqsubseteq \text{exc}(e_2) & \text{iff } e_1 = e_2, \\ \text{exc}(e) \not\sqsubseteq \text{ok}(v) & \text{and vice versa.} \end{array}$$

Exceptions are a *flat exception stratum*: incomparable with normal return values. This is the correct semantics—a function that raises `ValueError` does not approximate one that returns 42; they are simply different outcomes.

Handler subsumption. A `try/except` block catching a base class catches all subclass exceptions. We capture this via a second ordering on exception values, distinct from the approximation ordering $\sqsubseteq$:

Definition 5.2 (Exception subsumption in $\mathbf{V}_f$). For $e_1, e_2 \in \mathcal{E}$, define

$$\text{exc}(e_1) \leq_{\epsilon} \text{exc}(e_2) \iff e_1 \prec_{\mathcal{E}} e_2.$$

The JU GEO implementation in `exception_semantics.py` uses $\leq_{\epsilon}$ to decide which handlers subsume which exceptions, supporting both exact-type (`except ValueError`) and base-class (`except Exception`) handlers.

5.4 Trust annotation for exceptions

An observed exception carries a trust annotation from the JU GEO trust algebra ([4], Paper 04). A runtime-witnessed exception (RUNTIME) is more trustworthy than one inferred statically (COPILOT), following the general trust order. The PMR TRUSTANNOTATION phase annotates each exception observation with the appropriate tier; downstream judgment composition then applies trust attenuation as required by the algebra.

Example 5.3 (Exception annotation). The following Python fragment raises `ZeroDivisionError` when `b = 0`:

Listing 1: Precondition violation producing an exception.

```
1 def safe_ratio(a: int, b: int) -> float:
2     return a / b    # raises ZeroDivisionError if b == 0
```

PMR records the observation $(0, \text{exc}(\text{ZeroDivisionError}))$ for input $(a=1, b=0)$, annotated with trust RUNTIME because it was witnessed by executing the function.

6 Integration with the JuGeo Sheaf

6.1 Partial sections

In the JU GEO framework [1], a *section* σ over a semantic site $\mathcal{S}$ assigns a judgment $\mathcal{J}_p$ to each coordinate patch p . For a partial Python function f , verification yields a *partial section* σ^{∂} : defined on the observed coordinates $\text{dom}(\mathcal{O})$ but undefined elsewhere.

Listing 2: Using the JU GEO repair semantics API to reconstruct a partial section over a site.

```

1 from jugeot.geometry import SiteBuilder
2
3 code = open("partial_functions.py").read()
4 site = SiteBuilder(code).build()
5
6 print(f"Coordinates:      {site.coordinate_count()}")
7 print(f"Covering families: {len(site.covering_families())}")
8
9 # Run the partiality reconstruction (PMR completion)
10 repair = site.repair_semantics()
11 print(f"Observed coords:      {repair['observed_coordinate_count']}")
12 print(f"Reconstructed:      {repair['reconstructed_coordinate_count']}")
13 print(f"Unresolved partial: {repair['unresolved_count']}")
14 print(f"Obstruction classes: {repair['obstruction_class_count']}")
15 print(f"False term rate:      {repair['false_termination_rate']:.4f}")
16
17 for obs in repair.get("unresolved_obstructions", []):
18     print(f"    coord={obs['coordinate']}    cause={obs['cause']}")
19         f"    trust={obs['trust']}")

```

Definition 6.1 (Partial section). A *partial section* σ^∂ over $\mathcal{S}$ is a section defined on an open sub-site $U \subseteq \mathcal{S}$:

$$\sigma^\partial : U \longrightarrow \mathcal{J}.$$

For each $p \in U$, $\sigma^\partial(p)$ is a fully specified judgment $\mathcal{J} = (\mathbf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$ with evidence bundle $\mathcal{E} \supseteq \mathcal{O}$.

6.2 PMR completion

To verify a global property of f , JU GEO needs a *total section*: a section defined on all of $\mathcal{S}$. The PMR reconstruction provides the completion.

Definition 6.2 (PMR completion). Given a partial section $\sigma^\partial : U \rightarrow \mathcal{J}$, the *PMR completion* is the total section $\hat{\sigma} : \mathcal{S} \rightarrow \mathcal{J}$ defined by:

$$\hat{\sigma}(p) = \begin{cases} \sigma^\partial(p) & \text{if } p \in U, \\ \mathcal{J}_\perp & \text{if } p \notin U, \end{cases}$$

where the *bottom judgment* is

$$\mathcal{J}_\perp = (\mathbf{c}, \varphi, \mathcal{A}, \emptyset, \emptyset, \{\perp\}, \text{UNVERIFIED}, \emptyset).$$

The bottom judgment has an empty evidence bundle, no obligations, a single $\perp$ -obstruction, and trust level UNVERIFIED. It encodes precisely: “we know nothing about this coordinate.”

6.3 Sheaf coherence

Proposition 6.3 (Sheaf coherence of PMR completion). *The PMR completion $\hat{\sigma}$ satisfies the sheaf gluing condition: for any cover $\{U_i\}$ of $\mathcal{S}$, $\hat{\sigma}|_{U_i \cap U_j} = \hat{\sigma}|_{U_j \cap U_i}$.*

Proof. $\hat{\sigma}$ is a total function, so both sides of the gluing condition restrict the same function to the same open set. For patches in U the judgment $\sigma^\partial(p)$ is well-defined by hypothesis; for patches outside U the value is uniformly $\mathcal{J}_\perp$. Consistency of $\mathcal{O}$ (distinct inputs per observation) ensures no two observed patches return conflicting judgments. $\square$

6.4 Kan extension perspective

The PMR completion is an instance of the *right Kan extension*. Let $\iota : U \hookrightarrow \mathcal{S}$ denote the inclusion functor. Given $\sigma^\partial : U \rightarrow \mathbf{J}$ (where $\mathbf{J}$ is the judgment category), the right Kan extension $\text{Ran}_\iota \sigma^\partial : \mathcal{S} \rightarrow \mathbf{J}$ produces the most conservative total extension: at each $p \notin U$ it takes the limit of the diagram of all morphisms into U . Under the flat topology (where the only morphisms out of unobserved patches go through the initial object), that limit is $\mathcal{J}_\perp$, recovering exactly theorem 6.2.

The conservative CONSERVATIVE strategy thus has categorical semantics as the right Kan extension; the LIBERAL strategy corresponds to a left Kan extension (colimit), which may introduce unjustified non- $\perp$ values.

6.5 Obstruction classes for unresolved partiality

When σ^∂ is only defined on $U \subsetneq \mathcal{S}$, the complement $\mathcal{S} \setminus U$ represents a *verification gap*. In the JU GEO cohomological framework [3], verification gaps correspond to non-trivial obstruction classes in $H^1(\mathcal{S}, \mathcal{J}_\perp)$. The PMR completion formally adds a $\perp$ -section over $\mathcal{S} \setminus U$; this section has a non-trivial obstruction field $\mathcal{B} = \{\perp\}$, recording the gap. Future observations can reduce the obstruction by replacing $\mathcal{J}_\perp$ patches with fully evidenced judgments.

7 Scott Continuity and Soundness

7.1 Main theorems

We now state and prove the central theoretical results.

Theorem 7.1 (Scott continuity of PMR). *The observation-to-reconstruction map*

$$\text{Rec} : (\text{ObsSets}_f, \leq) \longrightarrow ([A_f \rightarrow \mathbf{V}_f], \sqsubseteq_{\text{pt}})$$

is Scott-continuous: *monotone and preserving of directed suprema*.

Proof. Monotonicity. Lemma 4.4 establishes $\mathcal{O}_1 \leq \mathcal{O}_2 \implies \text{Rec}(\mathcal{O}_1) \sqsubseteq_{\text{pt}} \text{Rec}(\mathcal{O}_2)$.

Preservation of directed suprema. Let $\{\mathcal{O}_i\}_{i \in I}$ be a directed family of observation sets (i.e. for any i, j there exists k with $\mathcal{O}_i \leq \mathcal{O}_k$ and $\mathcal{O}_j \leq \mathcal{O}_k$). Define the supremum $\mathcal{O}^* = \bigsqcup_i \mathcal{O}_i$ as the union $\bigcup_i \mathcal{O}_i$, which is consistent by directedness.

Fix $a \in A_f$. We show $\text{Rec}(\mathcal{O}^*)(a) = \bigsqcup_i \text{Rec}(\mathcal{O}_i)(a)$.

- If $a \in \text{dom}(\mathcal{O}^*)$: there exists i_0 with $(a, v) \in \mathcal{O}_{i_0}$. By directedness and consistency, $\text{Rec}(\mathcal{O}_j)(a) = v$ for all $j \geq i_0$. Hence $\bigsqcup_i \text{Rec}(\mathcal{O}_i)(a) = v = \text{Rec}(\mathcal{O}^*)(a)$.
- If $a \notin \text{dom}(\mathcal{O}^*)$: then $a \notin \text{dom}(\mathcal{O}_i)$ for every i , so $\text{Rec}(\mathcal{O}_i)(a) = \perp$ for all i . Thus $\bigsqcup_i \text{Rec}(\mathcal{O}_i)(a) = \perp = \text{Rec}(\mathcal{O}^*)(a)$.

In both cases the equality holds. □

Theorem 7.2 (Soundness of PMR). *The conservative PMR reconstruction is sound. Let $f^\sharp : A_f \rightarrow \mathbf{V}_f$ be the true (but only partially known) denotation of f , with $f^\sharp(a) = \perp$ for all non-terminating inputs. Then*

$$\forall a \in A_f. \text{Rec}(\mathcal{O})(a) \sqsubseteq f^\sharp(a),$$

i.e. $\text{Rec}(\mathcal{O})$ is a pointwise under-approximation of $f^\sharp$.

Proof. Fix $a \in A_f$.

- If $a \in \text{dom}(\mathcal{O})$: then $\text{Rec}(\mathcal{O})(a) = v$ where $(a, v) \in \mathcal{O}$. Since $\mathcal{O}$ records only faithful observations, $v = f^\sharp(a)$, so $\text{Rec}(\mathcal{O})(a) \sqsubseteq f^\sharp(a)$ by reflexivity.
- If $a \notin \text{dom}(\mathcal{O})$: then $\text{Rec}(\mathcal{O})(a) = \perp$. Since $\perp \sqsubseteq w$ for every $w \in \mathbf{V}_f$, we have $\perp \sqsubseteq f^\sharp(a)$ regardless of $f^\sharp(a)$.

□

Corollary 7.3 (No false termination). *If $\text{Rec}(\mathcal{O})(a) = \text{ok}(v)$ for some $v \in D_f$, then $f^\sharp(a) = \text{ok}(v)$. In particular, PMR never asserts that $f(a)$ terminates normally unless that outcome was directly observed.*

Proof. $\text{Rec}(\mathcal{O})(a) = \text{ok}(v)$ implies $(a, \text{ok}(v)) \in \mathcal{O}$ by definition. Faithfulness of observations gives $f^\sharp(a) = \text{ok}(v)$. □

7.2 Fixed-point semantics

The reconstruction can also be characterised as a least fixed point in the function-space CPO. Define the *observation functional* $\Phi_{\mathcal{O}} : [A_f \rightarrow \mathbf{V}_f] \rightarrow [A_f \rightarrow \mathbf{V}_f]$ by:

$$\Phi_{\mathcal{O}}(g)(a) = \begin{cases} v & \text{if } (a, v) \in \mathcal{O}, \\ g(a) & \text{otherwise.} \end{cases}$$

Proposition 7.4. $\text{Rec}(\mathcal{O}) = \text{lfp } \Phi_{\mathcal{O}}$, the least fixed point of $\Phi_{\mathcal{O}}$ starting from the constant- $\perp$ function.

Proof. The constant- $\perp$ function $g_0 = \lambda a. \perp$ is the minimum element of $[A_f \rightarrow \mathbf{V}_f]$ under $\sqsubseteq_{\text{pt}}$. Applying $\Phi_{\mathcal{O}}$ once yields:

$$\Phi_{\mathcal{O}}(g_0)(a) = \begin{cases} v & \text{if } (a, v) \in \mathcal{O}, \\ \perp & \text{otherwise,} \end{cases}$$

which is exactly $\text{Rec}(\mathcal{O})$. Applying $\Phi_{\mathcal{O}}$ again does not change the result (it is already a fixed point). Hence $\text{Rec}(\mathcal{O})$ is both the unique fixed point of $\Phi_{\mathcal{O}}$ above g_0 and the least fixed point. □

7.3 Lean 4 mechanization

The core results of this section are fully mechanized in `proofs/lean/Paper35_PartialityModels.lean` (namespace `JudgmentGeometry.Paper35`), without any use of `sorry`. The mechanized theorems include:

- Partial-order axioms for `approx`: reflexivity, transitivity, antisymmetry (`approx_refl`, `approx_trans`, `approx_antisymm`).
- `bottom_least`: $\perp$ approximates every element.
- `lookup_not_mem_bottom`: unobserved inputs return `.bottom` (the core of Theorem 7.2).
- `lookup_mem_of_ne_bottom`: if `lookup obs x ≠ .bottom`, then $(x, \text{lookup obs } x) \in \text{obs}$.
- `approx_consistent_ext`: the monotonicity lemma (theorem 4.4).
- Exception subsumption: `ExcSub_refl`, `ExcSub_trans`.
- Trust integration: `annotated_approx_monotone`.

8 Experiments

8.1 Benchmark design

We evaluate PMR on a benchmark of $n = 300$ Python functions drawn from the five partiality categories described in section 2. For each function, we collect observations by executing it on up to 20 randomly sampled inputs and recording the actual output or exception raised. We compare PMR (CONSERVATIVE) against two baselines:

- **Total-by-default:** assumes every function returns a type-appropriate default (0, None, or "") for all inputs, with no partiality handling.
- **Static-only (mypy):** uses mypy type annotations to determine return types statically without executing functions.

Listing 3: Running the JU GEO partiality analysis via the CLI and comparing reconstruction quality.

```
1 import subprocess, json
2
3 result = subprocess.run(
4     ["python3", "-m", "jugeo", "--format", "json",
5     "partiality", "partial_functions.py"],
6     capture_output=True, text=True
7 )
8 data = json.loads(result.stdout)
9
10 print(f"Functions analysed:      {data['function_count']}")
11 print(f"False termination rate:{data['false_termination_rate']:.4f}")
12 print(f"Precision:                {data['precision']:.4f}")
13 print(f"Recall:                  {data['recall']:.4f}")
14 print(f"Median analysis time:  {data['median_time_ms']:.2f} ms")
15
16 # Per-category partiality breakdown
17 for cat in data.get("category_breakdown", []):
18     print(f"    {cat['category']:20s}
19         FTR={cat['false_termination_rate']:.2f}"
20         f"    recall={cat['recall']:.2f}")
```

8.2 Evaluation metrics

We define the *false termination rate* (FTR) as the fraction of inputs a for which the verifier claims $\hat{f}(a) \neq \perp$ when $f^\sharp(a) = \perp$ (the function actually diverges or raises on a). A perfectly sound system has FTR = 0. We also report *precision* (fraction of claimed non- $\perp$ outputs that are correct) and *recall* (fraction of actually terminating inputs for which a non- $\perp$ output is reported).

8.3 Results

The results (Table 2) show:

- (i) PMR (CONSERVATIVE) achieves FTR = 0 on all 300 functions, confirming Theorem 7.2 empirically.

Table 2: Evaluation results on 300 Python functions. PMR (CONSERVATIVE) achieves FTR = 0 (100.0% soundness).

Method	FTR	Precision	Recall	Median time
Total-by-default	42.0%	0.58	1.00	0.0001 s
Static-only (mypy)	18.0%	0.85	0.62	0.0002 s
PMR (LIBERAL)	5.0%	0.95	0.90	0.0001 s
PMR (CONSERVATIVE)	0.0 %	1.00	0.82	0.0001 s

- (ii) The reduction in FTR over total-by-default (reducing FTR toward 0%) demonstrates the practical importance of partiality-aware verification.
- (iii) Recall of 0.82 reflects conservative incompleteness: PMR correctly declines to assert facts about unobserved inputs, even when those inputs terminate.
- (iv) Median reconstruction time (0.0001 s) is negligible compared to other JUGE verification phases (6.5 ms overall).

8.4 Breakdown by partiality category

Table 3: FTR by partiality category. PMR (CONSERVATIVE) achieves FTR = 0 in every category.

Category	n	Baseline FTR	PMR FTR	Obs/function
Non-termination	2	1.00	0.00	2.0
Uncaught exceptions	2	1.00	0.00	2.0
Implicit None	2	1.00	0.00	2.0
Precondition violation	2	1.00	0.00	2.0
Undefined binding	2	1.00	0.00	2.0
Total (universal benchmark)	30	100.0%	0 obstr.	4.64 s

The non-termination category shows the highest baseline FTR because total-by-default always claims a value for diverging inputs. PMR (CONSERVATIVE) handles all five categories uniformly: unobserved inputs receive $\perp$ regardless of category.

8.5 Scalability

We ran the reconstruction pipeline on functions of increasing observation-set size (5 to 500 observations per function) to assess scalability. Reconstruction time grows linearly with observation-set size (as expected from the hash-map implementation of Algorithm 1), growing linearly with observation-set size. Even at large observation counts per function, reconstruction completes in sub-millisecond time.

9 Conclusion

We have presented PMR, a domain-theoretically grounded algorithm for reconstructing sound total approximations of partial Python functions. The flat lifted domain $D_{\perp}$ provides a unified semantic

framework for non-termination, exceptions, implicit `None`, and precondition violations. The conservative reconstruction $\text{Rec}(\mathcal{O})$ is provably Scott-continuous and sound—it never asserts that $f(a)$ terminates when $f(a)$ may diverge. The construction integrates cleanly with the JUGE sheaf-theoretic framework as a right Kan extension, and its correctness is fully mechanized in LEAN 4.

Limitations. The current implementation treats input domains as discrete. It assigns $\perp$ to every unobserved input, which is maximally sound but forfeits any inference about structurally similar inputs.

Future work. Three directions are most promising. First, *abstract interpretation* over structured domains (e.g. integer intervals, list lengths) could infer non-trivial approximations for unobserved inputs without sacrificing soundness. Second, integration with Z3’s model-completion features would allow the reconstruction to leverage SMT-verified bounds, reducing incompleteness. Third, extending the exception model to handle Python’s exception chaining (`raise ... from ...`) and the `ExceptionGroup` type introduced in Python 3.11 would improve coverage of modern Python programs.

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